

Arthur Cavalcante de Andrade

COMPUTER SCIENCE · SOFTWARE ENGINEERING · AI RESEARCH

Goiânia, Brazil

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Summary

Software Engineer with 4+ years of professional experience, currently working with Python, Ruby on Rails, and PostgreSQL. Focused on clean code, performance, and automated testing, with extensive background architecting advanced AI-driven solutions leveraging LLMs, RAG architecture, and autonomous AI agents to automate complex workflows and document generation. Dual postgraduate specializations in Cybersecurity & Data Governance (PUC Minas) and IT Processes & Management (UFG), with hands-on experience in AppSec, ISO 31000 risk management, and enterprise Java Spring Boot.

I enjoy building solid, secure, and well-documented systems that bridge cutting-edge AI technologies with robust engineering practices to drive business value.

Work Experience

Federal Institute of Education, Science and Technology of Goiás (IFG)

Goiânia, Brazil - Remote

SOFTWARE ENGINEER

Jun. 2026 - Present

- Engineered and maintained scalable software components and digital services, actively participating in team meetings for product planning and prioritization;
- Executed the complex migration of the application from Ruby on Rails 5 to 8, ensuring system stability, dependency compatibility, and adherence to best practices;
- Restructured the frontend by building highly reactive, decoupled interfaces using ViewComponents and Hotwire;
- Improved code quality and deployment workflows by optimizing GitLab CI/CD pipelines and systematically resolving legacy technical debt.

Court of Justice of the State of Goiás

Goiânia, Brazil - Hybrid

LEAD SOFTWARE ENGINEER

May. 2026 - Present

- Leading the architectural migration of the MCA contract management system for the NCCA and the Kaizen strategic system from legacy stacks (Ruby on Rails/Node.js) to a unified Java Spring Boot backend;
- Directing the engineering team by establishing architectural standards, conducting code reviews, and translating stakeholder business requirements into scalable technical tasks;
- Enhancing application stability and performance through continuous bug resolution and optimization of algorithms and PostgreSQL database queries.
- Stack: Java (Spring Boot), TypeScript (React.js), PostgreSQL, GitLab.

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SOFTWARE ENGINEER | IT RISK MANAGER

Feb. 2025 - May. 2026

- Designed and developed web-based risk management applications and AI pipelines to automate data processing and streamline operational workflows;
- Led the full risk management process (identification, analysis, and mitigation) across IT projects, vendor management, and emerging technologies;
- Served as the primary liaison for compliance and third-party assessments, ensuring adherence to ISO 31000, LGPD, GDPR, and CNJ resolutions;
- Improved monitoring and strategic visibility by developing robust data dashboards tracking critical KPIs and KRIs.
- Stack: Ruby (Rails 8+, Hotwire), Python (FastAPI), TypeScript (React.js), PostgreSQL, Docker, GitHub.

Cilia Tecnologia (Insurtech)

Goiânia, Brazil - Hybrid

SOFTWARE ENGINEER

Feb. 2023 - Jan. 2026

- Designed and deployed high-traffic RESTful APIs in Ruby on Rails to support core business operations and external integrations;
- Built and maintained responsive user interfaces and screen flows using Vue.js and Backbone.js;
- Enhanced overall application performance by optimizing PostgreSQL database structures, automating routine tasks, and implementing efficient background workers;
- Championed code quality and engineering best practices through thorough code reviews, RSpec test automation, and detailed technical documentation.
- Stack: Ruby (Rails 3+), JavaScript (Vue.js, Backbone.js), Python, PostgreSQL, Bitbucket.

National Council for Scientific and Technological Development (CNPq)

Goiânia, Brazil - Hybrid

SCIENTIFIC INITIATION RESEARCHER

Sep. 2022 - Aug. 2023

- Researched advanced graph theory concepts, focusing on graphs with bounded maximum degrees;
- Formulated new mathematical theorems and conjectures by deeply analyzing the behavioral properties of cubic graphs;
- Engineered Python web scraping tools (BeautifulSoup) to systematically collect and represent real-world data as graph structures;
- Applied foundational algorithms and complex data structures to solve and optimize theoretical computational problems;
- Creating new theorems and conjectures for the research subject.

Education

Postgraduate Degree, IT Processes and Management FEDERAL UNIVERSITY OF GOIÁS	<i>Goiânia, Brazil</i> 2025 - 2027
Postgraduate Degree, Cybersecurity and Data Governance PUC MINAS	<i>Belo Horizonte, Brazil</i> 2024 - 2025
B.S., Computer Science FEDERAL UNIVERSITY OF GOIÁS	<i>Goiânia, Brazil</i> 2018 - 2024

Key Projects

Contracts and Acquisitions Module (MCA) COURT OF JUSTICE OF GOIÁS	<i>Goiânia - GO</i> Apr. 2026
Comprehensive planning and management system implemented at TJGO to streamline contract administration and payment workflows. Transitioned the institution away from scattered spreadsheets by automating processes across every procurement stage. Features the Risk Assessment Engine.	
Risk Assessment Engine COURT OF JUSTICE OF GOIÁS INTELLECTUAL PROPERTY (No. BR512026002418-8)	<i>Goiânia - GO</i> Feb. 2026
Automated risk assessment solution powered by a Python and FastAPI backend, integrating RAG architecture and NLP to translate qualitative public sector documents into quantitative risk metrics. Built upon the ISO 31000 methodology, the system leverages historical knowledge bases to calibrate probability indices, functioning as a highly scalable GRC tool that reduces total assessment time by 80%.	

Honors & Awards

HACKATHON	
2023	Award Winner , Managed a team of three high school students during a low-code hackathon at the 1st EPT Congress (CETT/UFG), leading the design and development of a mobile app (using MIT App Inventor) integrated with a prototype reservoir to collect and monitor rainwater runoff. The project received a podium award and was later incubated by CEI/UFG.
	<i>Goiânia, Brazil</i>

Skills

Languages & Frameworks	Ruby (Rails), Python (FastAPI, PyTorch), Java (Spring Boot), C/C++ (CUDA), JS (React/Vue).
AI & Data	RAG Architectures, LLMs, Vector Databases, Semantic Retrieval, Machine Learning.
DevOps & Tools	PostgreSQL, SQLite, Docker, Git (GitLab, GitHub, Bitbucket), Linux/Windows.
Security & Governance	SSDLC, AppSec, Burp Suite, ISO 31000, Data Governance (LGPD/GDPR), Security Auditing.
Methodologies	TDD, ITIL/COBIT, Algorithm Design, Cross-Functional Team Guidance.
Languages	Portuguese (Native), English (Proficient).

Certifications

2026	C1 Proficient , EF-SET
2024	Junior Cybersecurity Analyst , CISCO Networking Academy
2024	Network Technician , CISCO Networking Academy
2023	Fundamentals of Accelerated Computing with CUDA C/C++ , NVIDIA
2022	Data Privacy and Protection (LGPD) , SENAI